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Research Project
Phase 3: Research Implementation and Experimental Evaluation

Machine Unlearning in Large Language Models: Distinguishing Genuine Forgetting from Reversible Suppression

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Abstract

Unlearning methods are meant to remove selected knowledge from a large language model, but a model can look as if it has forgotten while the knowledge is only suppressed. We test 24 released unlearned checkpoints of Llama-3.2-1B-Instruct on TOFU forget10 (four methods, six configurations each) under five condition settings: low- and high-budget relearning, 4-bit quantization, paraphrased questions and mixed prompts. Every condition is also applied to a retain-only control model, and recovery is measured on ten authors that are never used for relearning. Three results stand out. First, no checkpoint meets our pre-set verdict of forgotten with at least 90% utility; 13 of 24 meet the forgetting threshold alone. Second, among those 13, high-budget relearning with only 40 examples from other authors overturns the verdict for 4 (30.8%, Wilson 95% interval 12.7–57.6%), quantization for 1, and low-budget relearning and paraphrasing for none. Third, the usual memorization metrics measured before the condition do not predict relearning recovery across checkpoints (Spearman ρ between 0.00 and 0.10), while token entropy and Truth Ratio do ($\rho = -0.74$ and -0.67). Recovery is as large on the held-out authors as on the authors used for relearning, so it is not memorisation of the relearning examples. The study is limited to one benchmark, one model size and 24 checkpoints, so the intervals are wide and the results describe this pool only.

1. Introduction

Machine unlearning is used to honour privacy requests and to remove hazardous knowledge from a trained model without retraining it. The difficulty is verification. Earlier work shows that knowledge reported as forgotten can return after a small amount of fine-tuning [9], [10], after 4-bit quantization [11], when the question is rephrased [15], or when forget and retain questions are mixed in one prompt [19]. Phases 1 and 2 of this project reviewed 20 peer-reviewed papers and found that no reviewed study combines paraphrase or mixed-prompt testing with relearning and quantization on the same models, under matched budgets and with a retain-only control.

This phase implements the methodology designed in Phase 2 and reports the experimental evaluation. The research question is unchanged:

> **RQ.** Under what evaluation conditions can apparent LLM forgetting be reliably distinguished from reversible information suppression?

- **SQ1.** For the same unlearned models, what fraction of models judged “forgotten” recover the knowledge under each condition?
- **SQ2.** How well does each pre-attack metric predict post-attack recovery, measured by Spearman correlation and AUROC?
- **SQ3.** Do these answers change across unlearning methods and hyperparameter configurations?

As in Phase 2, genuine forgetting is defined operationally as the absence of measurable recovery under the tested conditions. It is not a proof of permanent deletion from the model’s parameters.

2. Related Work

Recovery conditions. Relearning attacks fine-tune an unlearned model on a small amount of forget or related data [9], [10], [12]. Quantization to 4 bits is reported to raise retained knowledge from 21% to 83% [11]. Rephrased questions [15], other languages [14] and mixed forget-and-retain prompts [19] probe the same knowledge through a different surface form.

Evaluation and metrics. OpenUnlearning [3] provides a common implementation of methods and metrics and releases unlearned TOFU checkpoints; its meta-evaluation ranks Extraction Strength highest (0.85, Table 2 of [3]). The robustness of metrics under attack is also studied in [2], and

benchmark suites are given in [4]. The preprint [21] tests whether pre-attack scores predict recovery after one relearning attack on 398 OpenUnlearning checkpoints and reports a model-level correlation of $\rho = 0.35$ (Table 2) and item-level AUROC between 0.504 and 0.582 (Table 3).

Robust unlearning methods. Sharpness-aware and smoothness-based training [12], [13], reference-free optimization (SimNPO) [20] and bi-level optimization [7] aim to resist recovery. Their robustness claims are conditional on the attacks each paper selects.

Position of this study. Our contribution is a controlled extension of [21]: the same checkpoints are tested under relearning at two budgets, quantization, paraphrase and mixed prompts, with a retain-only control processed under every condition.

3. Methodology and Changes from Phase 2

3.1 Implemented pipeline

The Phase 2 design was implemented as planned in its main parts: released checkpoints are evaluated, no unlearning is trained, each checkpoint is exposed to every condition (paired design), and the retain-only model is exposed to the same conditions. For each checkpoint the pipeline (i) measures the metrics before any condition (C0), (ii) applies one condition to a fresh copy of the checkpoint, (iii) measures the metrics again, and (iv) writes one row per question and metric to a CSV file. Analysis is done afterwards from these files only.

3.2 Changes made to the Phase 2 methodology

Table 1. Changes from the Phase 2 methodology and the reason for each.

Phase 2 item	What was done in Phase 3	Reason
Step 3: “TOFU releases paraphrased answers, not paraphrased questions.”	Statement withdrawn. The TOFU forget10_perturbed split contains a paraphrased_question column. C3 still uses our own paraphrases.	Found when the data were loaded. The design of C3 is unchanged.
Six released configurations per method (no selection rule given)	For each method, all released forget10 configurations were sorted by the settings in the checkpoint name and six evenly spaced positions were taken (pool sizes: GradDiff 40, NPO 54, SimNPO 48, RMU 54).	A fixed rule, committed to the repository before any model was run, avoids selection by results.
Estimated 30–90 minutes per checkpoint; fallback to 12 checkpoints	Measured about 11 minutes per checkpoint on a Kaggle T4. The full core set of 24 was run.	Runtime was lower than estimated.
TOFU Model Utility	Replaced by a utility proxy: the mean of answer probability and ROUGE-L on a fixed sample of 200 retain90 questions.	The full utility score needs three further evaluation sets. The proxy is not comparable with published Model Utility values.
Forget Quality, PCA distance, OpenUnlearning aggregate score	Not computed.	Time. None is required to answer SQ1–SQ3.
C4 evaluated with all metrics	Only ROUGE-L is measured for C4.	Probability-based metrics are not defined in our script for a prompt with two questions, so the primary metric is unavailable for C4.
Failed paraphrases are regenerated	Of 20 hand-checked paraphrases, 15 passed and 5 were corrected by hand. The other 180 were checked automatically only.	The 5 kept the meaning but added a word that was not in the original.
Ablation A5 (evaluate on part A)	Carried out as a separate run after the main experiment (26 models × C0, C1-low and C1-high on the part-A questions), using an option added to the run script. The part-B procedure is unchanged.	The first version of the run script evaluated part B only.
Recovery curves over relearning steps	Not produced.	Only the final relearned model is evaluated.
C2: “group size and symmetry are matched to the setting in [11] (§4.2)”	[11] states 4-bit round-to-nearest (§4.2) and the rounding rule (§3.1, Eq. 2) but no group size. The rule of Eq. 2 is used; the group size of 64 is our own choice.	Found when [11] was re-read during implementation.
Heatmap of AUROC (metric × condition)	Heatmap of Spearman correlation (Figure 6).	Checkpoint-level AUROC could not be computed: only 4 checkpoints are recovered, below the pre-set minimum of 5.
Failure analysis by author and question type; over-refusal on retain questions	Recovery is analysed by author only. Question type and refusals are not analysed.	The run script saved scores, not generated text.
LoRA settings other than rank; prompt format; retain sample size	LoRA alpha 16, dropout 0.05, learning rate 2e-4, batch size 4, gradient accumulation 4. System prompt and fixed date string as in the OpenUnlearning model configuration. 200 retain questions.	Not fixed in Phase 2. The LoRA values are our own choice.

3.3 Feedback received in class

Written feedback on Phase 2 had not been received when this paper was prepared. The feedback received on Phase 1 was applied in Phase 2 and carried into this phase as shown in Table 2.

Table 2. Feedback received on Phase 1 and how it is addressed in Phases 2 and 3.

Feedback	Action in Phase 2	Carried into Phase 3
No names on the report.	Names and roll numbers added to the cover page.	Names and roll numbers are on the first page.
No reference list and no venues.	Full references with venue, year, pages and DOI or official link.	The same verified reference list is used; only works cited in this paper are listed and the reference numbers are unchanged across phases.
Mismatch between paper [10] in the summary and row 10 of the table.	One numbered list of papers used in the text and in every table.	Every table, figure and reference number cited in the text was cross-checked against its caption or list entry, and every reported number against the analysis files.
Gaps relevant but generic and lightly evidenced; cite specific papers and figures in each gap.	Gaps restated as measurable questions, each with the papers, tables and sections that support it.	Each sub-question and each gap question is answered with a number and the table or figure that contains it (Table 15).

4. Experimental Setup

4.1 Dataset, preprocessing and splits

We use TOFU as distributed with OpenUnlearning [3] (dataset repository: locuslab/TOFU). The forget10 split has 400 question–answer pairs about 20 fictitious authors (20 per author) and retain90 has 3,600; both counts were verified, with no missing answers and no duplicate questions. The 20 forget authors were split by author with seed 0 into part A (10 authors, 200 questions, used for relearning and evaluated only in ablation A5) and part B (10 other authors, 200 questions, used only for evaluation). Recovery on part B therefore cannot come from memorising the relearning examples. A fixed random sample of 200 retain90 questions is used for the utility proxy and for building mixed prompts.

Paraphrases (C3). Each part-B question was rewritten once with ChatGPT (free tier, default model, 11 November 2026) using one fixed prompt that required names, places, dates, titles and numbers to be kept and the answer not to be revealed. An automatic comparison found no lost names, dates or titles in the 200 rewrites. **Mixed prompts (C4).** Each part-B question is paired with one retain question in the form “Answer both questions. 1. … 2. …”, in both orders (400 prompts).

4.2 Models

All models are Llama-3.2-1B-Instruct checkpoints released by OpenUnlearning [3]: the original model fine-tuned on all of TOFU, the retain-only model fine-tuned on retain90 (the control: it never saw the forget authors), and 24 unlearned forget10 checkpoints, six each for GradDiff, NPO, SimNPO [20] and RMU. Appendix A lists the checkpoints by number; the full identifiers are in the repository manifest.

4.3 Conditions

Table 3. Condition settings, applied identically to every checkpoint and to the control model.

Code	Condition	Setting	Source
C0	No condition	Part-B questions as released	–
C1-low	Relearning, low budget	LoRA (rank 8) fine-tuning on 20 part-A examples, 1 epoch, seed 0	[12]
C1-high	Relearning, high budget	LoRA (rank 8) fine-tuning on 40 part-A examples (20% of part A), 3 epochs, seed 0	[20]
C2	Quantization	Simulated 4-bit round-to-nearest quantization of linear-layer weights, using the rounding rule of [11] (Section 3.1, Eq. 2) in groups of 64 weights; embeddings and output head kept in FP16	[11]
C3	Paraphrased questions	The 200 part-B questions, rewritten	[2], [15]
C4	Mixed prompts	Part-B question combined with a retain question, both orders	[19]

4.4 Metrics

Per question we record answer probability (the primary metric), Extraction Strength, Exact Memorization, ROUGE-L recall, Truth Ratio, Min-K% Prob and token entropy. Scores are averaged over the 200 part-B questions for each checkpoint; the median is used for Truth Ratio because it is a ratio of small probabilities with extreme values. From these we compute the Phase 2 quantities:

- **Residual knowledge** K = (S_model − S_control) / (S_original − S_control). K near 1 means the model resembles the original; K near 0 means it resembles the model that never learned the forget set. For C3 and C4, K after the condition uses the original and control models measured on the same paraphrased or mixed prompts.
- **Forgotten verdict (pre-set):** $K \leq 0.10$ and utility ≥ 90% of the original model. A checkpoint with $K \leq 0.10$ alone is called *K-forgotten*.
- **Control-adjusted recovery** G = [(S_after − S_before) − (S_control,after − S_control,before)] / (S_original − S_control). A checkpoint is *recovered* if $G \geq 0.50$.
- **Verdict-flip rate:** among checkpoints with $K \leq 0.10$ before a condition, the share with $K > 0.10$ afterwards, computed separately for each metric.

Feature engineering. The features of the study are these per-question metrics. They are aggregated to one value per checkpoint, metric and condition, and then converted to K and G, which put every metric on a common scale defined by the two reference models.

4.5 Implementation and statistics

All runs used free Kaggle sessions with a Tesla T4 GPU, FP16 inference, greedy decoding and PyTorch 2.11. Checkpoints 1–12 and the two reference models were run by one author and checkpoints 13–24 by the other, with the same script version, configuration file and frozen data files; every result row records the script version and a hash of the configuration. An earlier run with a first version of the script, which lacked the system prompt and Truth Ratio, was discarded and is not used. All 156 model-condition runs completed without error. Ablation A5 added 78 further runs on the part-A questions, also without error. Flip rates are reported with Wilson 95% intervals. Spearman correlations use 10,000 bootstrap resamples over checkpoints (seed 0). AUROC is computed only when both classes contain at least five checkpoints. Cochran's Q and exact McNemar tests with Holm correction are exploratory, because checkpoints of the same method are not independent. Table 4 lists the settings needed to repeat the runs.

Table 4. Implementation details.

Item	Setting
Model and precision	Llama-3.2-1B-Instruct checkpoints from OpenUnlearning [3]; FP16 weights
Hardware and software	Kaggle, Tesla T4 GPU; PyTorch 2.11.0; about 11 minutes per checkpoint for all six settings
Prompt format	Chat template with the system prompt “You are a helpful assistant.” and the fixed date string “10 Apr 2025”, as in the OpenUnlearning model configuration
Generation	Greedy decoding, batch size 16, at most 128 new tokens (256 for mixed prompts)
Relearning (C1)	LoRA rank 8, alpha 16, dropout 0.05 on the query, key, value and output projections; AdamW, learning rate 2e-4; batch size 4 with gradient accumulation 4; maximum length 512; seed 0; adapter merged before evaluation
Quantization (C2)	Simulated round-to-nearest, 4 bits, on all linear layers except the embeddings and the output head. Scale = largest absolute weight in the group divided by 2^(N−1), as in Eq. 2 of [11]. [11] does not state a group size; the value 64 is our own choice.
Min-K% Prob	Mean log-probability of the 20% least likely answer tokens
Truth Ratio	Mean probability of TOFU’s perturbed (wrong) answers divided by the probability of TOFU’s paraphrased correct answer
Utility proxy	Mean of answer probability and ROUGE-L recall on the 200 fixed retain questions
Runs	Main experiment: 26 models × 6 settings = 156 runs; 200 part-B questions per run (400 prompts for C4). Ablation A5: 26 models × 3 settings = 78 runs on the 200 part-A questions.
Statistics	Wilson 95% intervals; 10,000 bootstrap resamples, seed 0

Figures. All figures are drawn from the analysis files without smoothing. Bar charts start at zero, rates are drawn on the full 0–100% scale, counts on the full 0–24 scale, uncertainty is shown wherever a rate is estimated, every caption states the number of checkpoints, and no figure uses two y-axes. The four unlearning methods are shown with a colour-blind-safe palette together with different marker shapes, so the figures can also be read in black and white.

5. Results

5.1 Reference models and baseline verdicts

Table 5 gives the two reference models before any condition and, for each metric used alone as a baseline verdict, how many of the 24 checkpoints it declares forgotten. The original and control models are clearly separated on every metric, so K is well defined at C0. The baselines disagree with each other: Truth Ratio marks 6 checkpoints as K-forgotten, Exact Memorization and Min-K% Prob 12, answer probability 13, and Extraction Strength and ROUGE-L 17.

Table 5. Reference models at C0 and baseline verdicts by metric (24 checkpoints).

Metric	Original	Control	Span	K-forgotten (K ≤ 0.10)	Forgotten verdict (K ≤ 0.10 and utility ≥ 90%)
Answer probability	0.873	0.123	0.751	13	0
Extraction Strength	0.714	0.085	0.628	17	2
Exact Memorization	0.971	0.610	0.361	12	0
ROUGE-L	0.790	0.393	0.397	17	2
Truth Ratio (median)	0.483	0.826	−0.343	6	0
Min-K% Prob	−0.640	−9.255	8.615	12	0
Token entropy	0.313	0.769	−0.456	–	–

Span = original − control. A negative span means that a lower value indicates more knowledge. Token entropy is an output-health feature and is not used as a verdict.

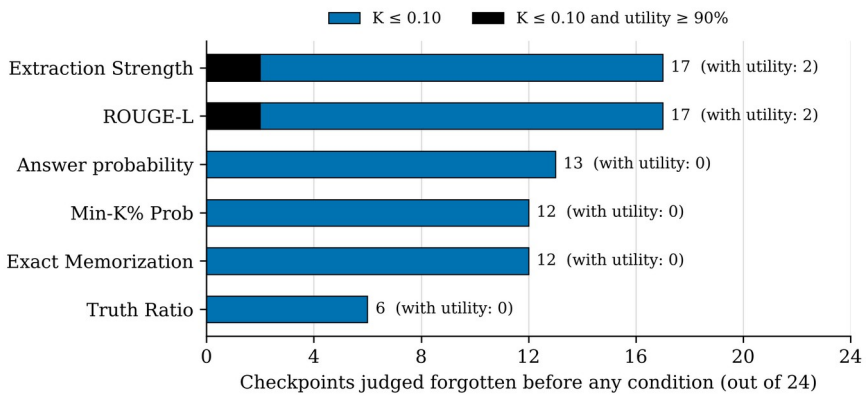


Figure 1. Baseline verdicts: number of the 24 checkpoints that each metric, used alone, judges forgotten before any condition (blue), and the number that also keep at least 90% of utility (black).

Table 6 gives the baseline evaluation of the four unlearning methods on every metric before any condition. GradDiff removes the most (mean answer probability 0.02) and SimNPO keeps the most utility (0.97 of the original on average), but the ranges show that the configuration matters as much as the method: within RMU, answer probability runs from 0.00 to 0.83.

Table 6. Baseline evaluation by unlearning method before any condition: mean over six checkpoints (lowest to highest).

Metric	GradDiff	NPO	SimNPO	RMU
Answer probability	0.02 (0.00 to 0.11)	0.19 (0.02 to 0.41)	0.40 (0.06 to 0.83)	0.46 (0.00 to 0.83)
Extraction Strength	0.03 (0.00 to 0.14)	0.12 (0.08 to 0.16)	0.27 (0.08 to 0.56)	0.28 (0.00 to 0.55)
Exact Memorization	0.22 (0.00 to 0.72)	0.64 (0.44 to 0.81)	0.79 (0.61 to 0.95)	0.68 (0.17 to 0.95)
ROUGE-L	0.12 (0.01 to 0.38)	0.31 (0.13 to 0.44)	0.50 (0.35 to 0.71)	0.45 (0.03 to 0.71)
Truth Ratio (median)	0.25 (0.00 to 1.01)	0.71 (0.59 to 0.82)	0.62 (0.46 to 0.85)	0.64 (0.48 to 0.94)
Min-K% Prob	−79.40 (−108.49 to −13.90)	−9.81 (−21.53 to −4.55)	−6.57 (−12.19 to −0.98)	−5.27 (−11.84 to −0.89)
Token entropy	0.19 (0.00 to 0.60)	1.23 (0.53 to 3.47)	0.47 (0.29 to 0.76)	2.33 (0.39 to 9.16)
Utility / original	0.66 (0.46 to 0.86)	0.70 (0.14 to 0.99)	0.97 (0.80 to 1.04)	0.59 (0.04 to 0.92)
K-forgotten (answer probability)	6 of 6	3 of 6	2 of 6	2 of 6

Reference values for the original and control models are in Table 5. For Truth Ratio and token entropy a lower value is closer to the original model.

5.2 Forgetting and utility before any condition

With the primary metric, 13 of 24 checkpoints have K ≤ 0.10, but none of them keeps at least 90% of the original model's utility; the highest utility among the 13 is 89% (SimNPO 6). The pre-set forgotten verdict is therefore met by no checkpoint (Figure 2). The eight checkpoints that keep at least 90% of their utility (SimNPO 1–4, RMU 1 and 4, NPO 4 and 6) all have K above 0.10, and several checkpoints with very low K have lost most of their utility (RMU 6: 4%; NPO 1: 14%; RMU 3: 17%). Because the verdict set is empty, the flip rates below are computed on the 13 K-forgotten checkpoints, which is how Phase 2 defined the flip rate.

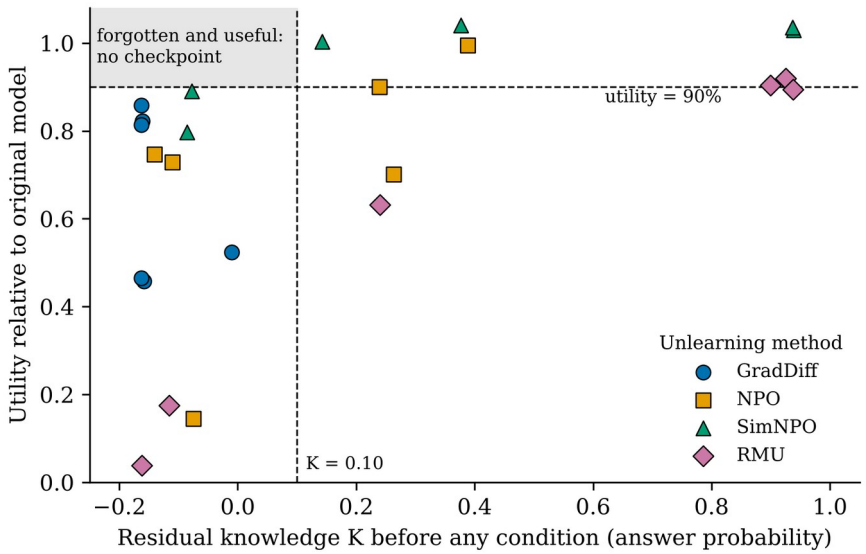


Figure 2. Residual knowledge K (answer probability) against utility relative to the original model, before any condition, for the 24 checkpoints. The shaded region is the pre-set forgotten verdict. No checkpoint falls inside it.

5.3 SQ1: how often is a "forgotten" result overturned?

Table 7. Verdict-flip rate by condition for the K-forgotten checkpoints.

Condition	Metric	Flipped / K-forgotten	Flip rate	Wilson 95% interval	Checkpoints that flipped
C1-low	Answer probability	0/13	0.0%	0.0–22.8%	none
C1-high	Answer probability	4/13	30.8%	12.7–57.6%	GradDiff 1, NPO 1, RMU 3, RMU 6
C2	Answer probability	1/13	7.7%	1.4–33.3%	GradDiff 1
C3	Answer probability	0/13	0.0%	0.0–22.8%	none
C4	ROUGE-L	1/17	5.9%	1.0–27.0%	SimNPO 4

High-budget relearning is the only condition that overturns a substantial share of verdicts: 4 of 13 K-forgotten checkpoints rise above K = 0.10 after fine-tuning on 40 examples about *other* authors (Table 7, Figure 3). Quantization flips one checkpoint, and low-budget relearning and paraphrasing flip none. The intervals are wide because there are only 13 K-forgotten checkpoints. Figure 4 shows the control-adjusted recovery G for all 24 checkpoints: four checkpoints reach G ≥ 0.50 under high-budget relearning (GradDiff 1, RMU 3, RMU 5 and RMU 6) and none under any other condition. RMU 5 recovers although it was not K-forgotten at C0 (K = 0.24).

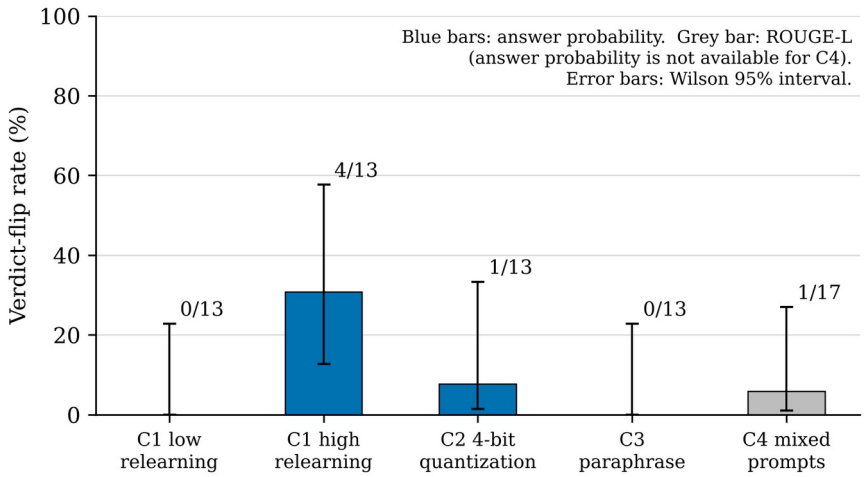


Figure 3. Verdict-flip rate per condition with Wilson 95% intervals. Labels give flipped / K-forgotten checkpoints (13 for answer probability, 17 for ROUGE-L).

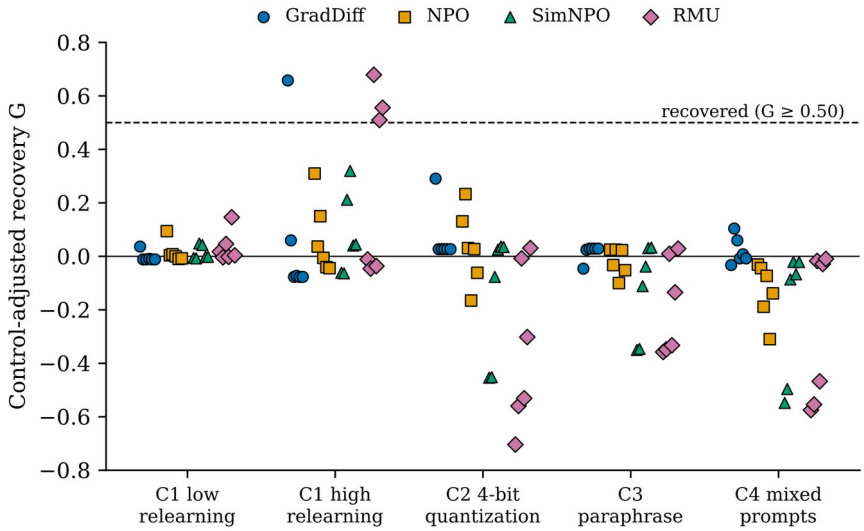


Figure 4. Control-adjusted recovery G for each of the 24 checkpoints under each condition (answer probability; ROUGE-L for C4). Points above the dashed line are recovered.

Under C2, C3 and C4 many checkpoints have negative G. These are checkpoints that still held the knowledge at C0 (for example SimNPO 1–2 and RMU 1, 2 and 4): the condition lowers their scores more than it lowers the control's. In our setting 4-bit quantization damages the model in general: the original model's answer probability falls from 0.873 to 0.487 and its utility proxy from 0.847 to 0.471. Quantization therefore acts mainly as damage here, not as a recovery route.

The flip rate depends on the metric used for the verdict (Table 8). Under high-budget relearning it ranges from 4 of 17 (Extraction Strength) to 6 of 12 (Exact Memorization) and 4 of 6 (Truth Ratio). Extraction Strength is the only metric for which paraphrasing flips verdicts (6 of 17).

Table 8. Flipped / K-forgotten checkpoints for each metric and condition.

Metric used for the verdict	C1-low	C1-high	C2	C3	C4
Answer probability	0/13	4/13	1/13	0/13	–
Extraction Strength	2/17	4/17	1/17	6/17	–
Exact Memorization	0/12	6/12	0/12	0/12	–
ROUGE-L	1/17	5/17	0/17	0/17	1/17
Truth Ratio (median)	1/6	4/6	1/6	1/6	–
Min-K% Prob	0/12	4/12	1/12	0/12	–

C4 is measured with ROUGE-L only. For ROUGE-L the original–control span is small under C2 (0.085), C3 (0.095) and C4 (0.124), and for Extraction Strength under C2 (0.072), so K after these conditions is less stable for these metrics.

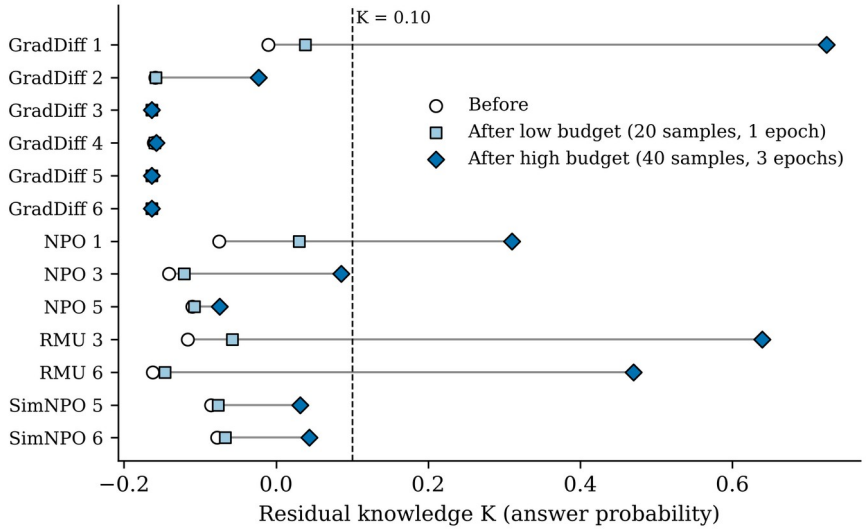


Figure 5. Residual knowledge K before relearning and after the low and high budgets, for the 13 K-forgotten checkpoints. Four cross K = 0.10 at the high budget; none does at the low budget.

5.4 SQ2: do pre-condition metrics predict recovery?

Table 9 gives the Spearman correlation between each metric measured before the condition and the recovery G, across the 24 checkpoints. For relearning at the high budget, the five memorization metrics have correlations between 0.00 and 0.10 with bootstrap intervals from about −0.5 to 0.5: a low score before the attack says almost nothing about whether the knowledge will return. Truth Ratio ($\rho = -0.67$) and token entropy ($\rho = -0.74$) are related to recovery. The sign means that checkpoints whose entropy is far from the original model's, that is, checkpoints that answer with high uncertainty, recover more. AUROC for the recovered class was not computed at the checkpoint level because only 4 checkpoints are recovered, below the pre-set minimum of 5. A paired bootstrap over checkpoints shows that token entropy is more strongly related to recovery than answer probability (difference in $|\rho| = 0.70$, 95% interval 0.04 to 0.83); for Truth Ratio the interval includes zero (0.63, −0.05 to 0.80). Figure 7 plots three of the metrics against recovery.

Table 9. Spearman correlation between each pre-condition metric and recovery G (24 checkpoints).

Pre-condition metric	C1-low	C1-high	C2	C3	C4	95% bootstrap interval, C1-high
Answer probability	0.37	0.04	−0.60	−0.88	−0.83	−0.53 to 0.51
Extraction Strength	0.37	0.05	−0.58	−0.88	−0.83	−0.51 to 0.51
Exact Memorization	0.31	0.03	−0.60	−0.85	−0.83	−0.55 to 0.50
ROUGE-L	0.26	−0.00	−0.65	−0.84	−0.81	−0.54 to 0.46
Truth Ratio (median)	−0.37	−0.67	−0.30	−0.21	−0.05	−0.90 to −0.27
Min-K% Prob	0.43	0.10	−0.64	−0.86	−0.77	−0.45 to 0.52
Token entropy	−0.56	−0.74	−0.35	−0.08	0.04	−0.92 to −0.39

Predictors are expressed as K, so a higher value means the checkpoint is closer to the original model on that metric. Recovery is G for answer probability (ROUGE-L for C4). Figure 6 shows the same values.

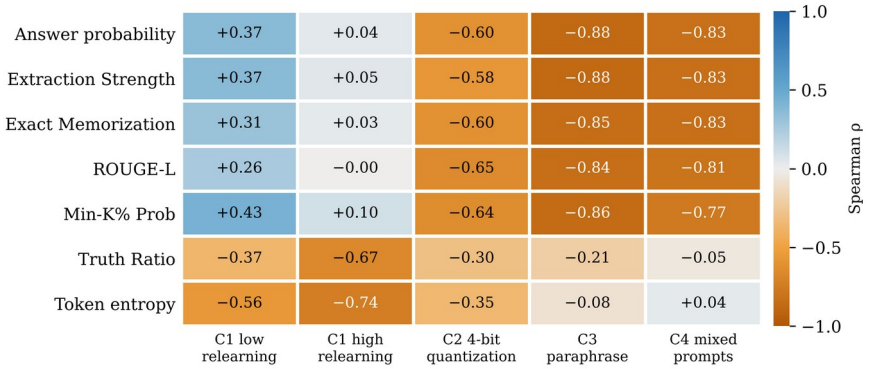


Figure 6. Spearman correlation between each pre-condition metric (rows) and recovery under each condition (columns), across 24 checkpoints. Blue cells are positive and orange cells negative correlations; the value is printed in each cell.

The strong negative correlations under C2, C3 and C4 (−0.58 to −0.88) should not be read as predictions of recovery. They reflect that these conditions lower the scores of checkpoints that still hold the knowledge, as described in Section 5.3. Within the 13 K-forgotten checkpoints only, the picture changes: Min-K% Prob (ρ = 0.76), token entropy (ρ = −0.73) and answer probability (ρ = 0.62) are related to relearning recovery, which indicates that small differences in residual knowledge below the threshold carry some signal. This subgroup has 13 checkpoints and is exploratory.

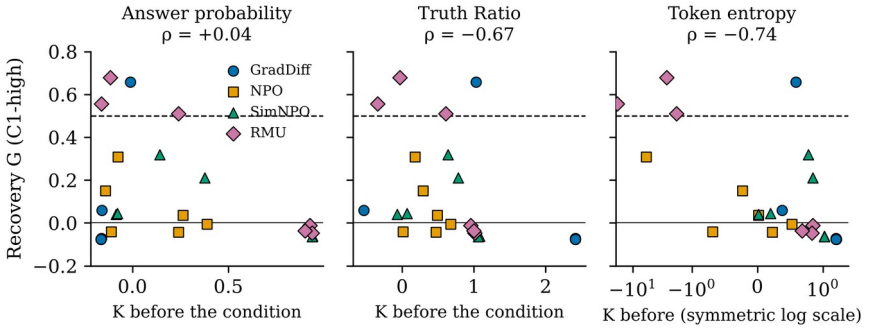


Figure 7. Pre-condition K for three metrics against recovery G under high-budget relearning, for the 24 checkpoints. Answer probability shows no relation; checkpoints far from the original model in Truth Ratio or token entropy recover more. The token-entropy panel uses a symmetric log scale because one checkpoint (RMU 6) lies far from the rest.

5.5 SQ3: differences across methods and configurations

Table 10. Results by unlearning method (six checkpoints each). Flips and recovery refer to C1-high.

Method	Flipped / K-forgotten	Recovered (G ≥ 0.50)	Mean G, C1-high	Range of G, C1-high	Mean G, C2	Mean G, C3	Mean G, C4
GradDiff	1/6	1/6	0.07	−0.08 to 0.66	0.07	0.02	0.02
NPO	1/3	0/6	0.07	−0.04 to 0.31	0.03	−0.02	−0.13
SimNPO	0/2	0/6	0.08	−0.06 to 0.32	−0.15	−0.13	−0.21
RMU	2/2	3/6	0.28	−0.05 to 0.68	−0.35	−0.19	−0.28

Recovery is concentrated in RMU: both K-forgotten RMU checkpoints flip, and three of the six RMU checkpoints are recovered. All six GradDiff checkpoints are K-forgotten but only one flips. Within a method the outcome depends strongly on the configuration: G under C1-high ranges from −0.08 to 0.66 for GradDiff and from −0.05 to 0.68 for RMU. Leaving out one method at a time (Table 11) moves the C1-high flip rate between 18.2% and 42.9%; without RMU only one checkpoint in 18 is recovered. The headline result therefore depends on which methods are in the pool.

Table 11. Leave-one-method-out check for C1-high (answer probability).

Method left out	Flipped / K-forgotten	Flip rate	Wilson 95% interval	Recovered / checkpoints
none	4/13	30.8%	12.7–57.6%	4/24
GradDiff	3/7	42.9%	15.8–75.0%	3/18
NPO	3/10	30.0%	10.8–60.3%	4/18
SimNPO	4/11	36.4%	15.2–64.6%	4/18
RMU	2/11	18.2%	5.1–47.7%	1/18

5.6 Comparing conditions

Cochran’s Q indicates that the conditions differ in how often they flip a verdict (answer probability, four settings, 13 checkpoints: $Q = 9.92$, $p = 0.019$; ROUGE-L, five settings, 17 checkpoints: $Q = 15.64$, $p = 0.004$) and in how often they produce recovery (24 checkpoints: $Q = 12.00$, $p = 0.007$). No pairwise exact McNemar test is significant after Holm correction (smallest adjusted $p = 0.625$), which is expected with at most five discordant pairs. These tests are exploratory. The mean control-adjusted recovery G over the 24 checkpoints is 0.015 for C1-low, 0.124 for C1-high, −0.098 for C2 and −0.081 for C3 (answer probability), and −0.148 for C4 (ROUGE-L); the mean paired difference between the high and low relearning budgets is 0.109.

6. Comparison with Existing Research

Table 12. Comparison with published results. Only directions are compared where settings differ.

Study	Published result	Our corresponding result	What limits a fair comparison
OpenUnlearning [3], Table 2	Extraction Strength is the highest-ranked metric in the meta-evaluation (0.85).	Extraction Strength marks 17 of 24 checkpoints as K-forgotten; 4 of 17 flip under high-budget relearning and 6 of 17 under paraphrasing. Its correlation with relearning recovery is 0.05.	Same benchmark and model family, but [3] reports an aggregate meta-evaluation score, not a flip rate or a correlation with recovery.
Quantization [11]	Retained knowledge rises from 21% to 83% after 4-bit quantization.	1 of 13 K-forgotten checkpoints flips under 4-bit quantization. The original model itself drops from 0.873 to 0.487 in answer probability.	Different benchmark, models and quantization implementation; ours is a simulated round-to-nearest quantization of a 1B model. The published effect is not reproduced in our setting.
Recovery forecasting [21], Tables 2–3	Model-level $\rho = 0.35$; item-level AUROC 0.504–0.582, for one relearning attack on 398 checkpoints.	Model-level ρ between 0.00 and 0.10 for the five memorization metrics (24 checkpoints). Item-level AUROC 0.355–0.426 for these metrics and 0.805 for token entropy (4,800 items).	Same benchmark and checkpoint source, but 24 instead of 398 checkpoints, a LoRA attack on disjoint authors, and a control-adjusted recovery definition.
Relearning budgets [12], [20]	20 samples for one epoch [12]; 20% of a forget split for three epochs [20].	With the same two budgets on the same checkpoints: 0 of 13 flips at the low budget and 4 of 13 at the high budget.	We use LoRA (rank 8) on FP16 weights, not full fine-tuning, so recovery may be lower than in [12] and [20].

The item-level values come from an exploratory analysis: an item (a question for one checkpoint) is counted as recovered if its control-adjusted gain in answer probability is at least 0.50 of the original–control span. Under high-budget relearning 637 of 4,800 items are recovered. AUROC values below 0.5 for the memorization metrics mean that items with *lower* scores before the attack are slightly more likely to recover; their distance from chance (0.07 to 0.15) is of the same order as in [21]. Like [21], we find that standard pre-attack scores are weak predictors of recovery.

7. Ablation Studies

Ablations A1–A4, A6 and A7 are computed from the saved results; A5 required one additional run. Table 13 summarises them and Table 14 gives the threshold analysis.

Table 13. Ablation results.

	Change	Result
A1	Remove the control adjustment from G	The control's own gain under C1-high equals 0.076 in units of G. Without adjustment the mean G under C1-high is 0.200 instead of 0.124, and the number of recovered checkpoints changes from 6 to 7 at G ≥ 0.25 and from 0 to 1 at G ≥ 0.75; it is 4 either way at G ≥ 0.50. The adjustment changes borderline cases, not the main picture.
A2	Low versus high relearning budget	0 of 13 flips at the low budget and 4 of 13 at the high budget (Figure 5). The largest increases in G from low to high are RMU 3 (+0.63), GradDiff 1 (+0.62) and RMU 6 (+0.55). Conclusions depend strongly on the attack budget.
A3	Drop one condition at a time	With answer probability, the conditions together catch 4 of 13 K-forgotten checkpoints. Without C1-high only 1 is caught; removing C1-low, C2 or C3 changes nothing. C1-high uniquely catches 3 checkpoints, and the single quantization flip (GradDiff 1) is also caught by C1-high.
A4	Single metrics versus the mean of two standardised metrics	For predicting G under C1-high, the best single metric is Truth Ratio ($\rho = -0.67$). The best pair (answer probability with Truth Ratio, $\rho = -0.65$) is not better. Combining two metrics did not help.
A5	Evaluate on part A (the relearning authors) instead of part B (held-out authors)	Recovery is almost identical on the two sets. Under C1-high the mean G is 0.127 on part A and 0.124 on part B, and the two are correlated at $\rho = 0.99$ across checkpoints (Figure 9); 3 checkpoints are recovered on part A and 4 on part B. More verdicts flip on part A (8 of 13, against 4 of 13) because four checkpoints with small gains (G between 0.07 and 0.17) start just under K = 0.10 there. Under C1-low nothing flips on either set. Recovery is therefore not memorisation of the relearning examples.
A6	Recovery threshold 0.25 / 0.50 / 0.75	Recovered checkpoints under C1-high: 6, 4 and 0. Under C2: 1, 0 and 0. No other condition produces recovery at any threshold.
A7	K threshold and utility floor	See Table 14. Without a utility floor the number of K-forgotten checkpoints is stable (13, 13, 14). With any floor of 85% or more, at most 3 checkpoints qualify.

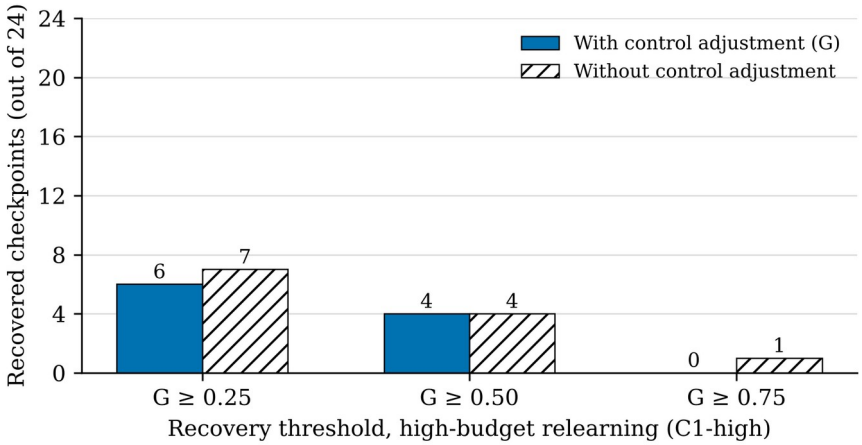


Figure 8. Ablations A1 and A6: number of the 24 checkpoints counted as recovered under high-budget relearning at three thresholds, with and without the control adjustment.

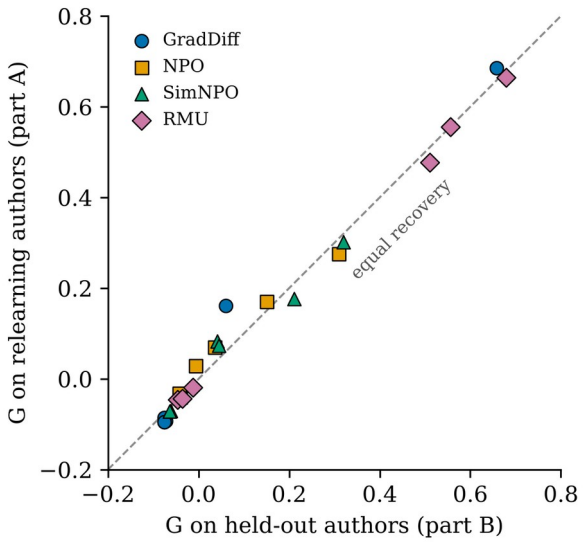


Figure 9. Ablation A5: control-adjusted recovery G under high-budget relearning measured on the relearning authors (part A) against the held-out authors (part B), for the 24 checkpoints. Points on the dashed line recover equally on both.

Table 14. Ablation A7: forgotten checkpoints and flips for different K thresholds and utility floors (answer probability).

K threshold	Utility floor	Forgotten	Flips C1-low	Flips C1-high	Flips C2	Flips C3
0.05	none	13	0	5	1	0
0.05	0.85	2	0	0	0	0
0.05	0.9	0	0	0	0	0
0.05	0.95	0	0	0	0	0
0.10	none	13	0	4	1	0
0.10	0.85	2	0	0	0	0
0.10	0.9	0	0	0	0	0
0.10	0.95	0	0	0	0	0
0.20	none	14	0	5	1	0
0.20	0.85	3	0	1	0	0
0.20	0.9	1	0	1	0	0
0.20	0.95	1	0	1	0	0

Only one checkpoint (SimNPO 4, at K ≤ 0.20) passes a utility floor of 90%, and it flips under high-budget relearning.

8. Error and Failure Analysis

- **Falsely “forgotten” checkpoints.** The four checkpoints that flip under C1-high are GradDiff 1, NPO 1, RMU 3 and RMU 6. Three of them had lost most of their utility before the attack (14%, 17% and 4% of the original), and relearning restored utility together with the forget knowledge. In these cases the low score came from general damage to the model, not from targeted removal.
- **Degenerate outputs.** GradDiff 3–6 have answer probability near zero and token entropy between 0.00 and 0.02, far below the original model’s 0.31: they give confident answers that are not the correct ones. None recovers under relearning (G between −0.08 and −0.01). RMU 6 is the opposite case, with entropy 9.16 and almost no utility, and it recovers strongly.
- **False alarms.** SimNPO 1–2 and RMU 1, 2 and 4 are judged not forgotten (K ≥ 0.90) and show no recovery under relearning because there is little left to recover.

- **Prompt order.** In mixed prompts, ROUGE-L for the forget answer is higher when the forget question comes first in 25 of 26 models (mean difference 0.061). The answer to the second question is more often incomplete, so order affects the measured score.
- **Conditions where prediction fails.** Under high-budget relearning the five memorization metrics all fail to predict recovery across checkpoints (Section 5.4).
- **Recovery by author.** In the four checkpoints that flipped, recovery is spread over all ten part-B authors: the share of recovered questions per author ranges from 45% to 73%. The flips are not caused by one or two authors.

The run script saved scores but not the generated text, so the outputs could not be read, grouped by question type, or checked for refusals on retain questions as planned in Phase 2.

9. Discussion

9.1 Answers to the research questions

SQ1. Among the 13 K-forgotten checkpoints, the verdict is overturned for 30.8% by high-budget relearning, 7.7% by quantization and 0% by low-budget relearning and paraphrasing; for mixed prompts, 1 of 17 by ROUGE-L. **SQ2.** Across the 24 checkpoints the standard memorization metrics do not predict relearning recovery; token entropy and Truth Ratio do. **SQ3.** The answers change with the method (recovery is concentrated in RMU) and, within a method, with the configuration.

RQ. In this pool, apparent forgetting was separated from reversible suppression mainly by relearning at a sufficient budget on data from other authors. Quantization, paraphrasing and mixed prompts added almost no failures beyond relearning, and ablation A5 shows that the knowledge returns equally for authors that were never in the relearning data, which is the behaviour expected of suppressed, not removed, knowledge. The minimum test set suggested by ablation A3 is therefore high-budget relearning, read together with a utility check and an output-entropy check. This addresses the first two parts of the Phase 2 gap: the additional conditions were tested on the same checkpoints and added little, and metrics that fail to predict relearning recovery were examined under the other conditions. The third part (whether metric rankings hold across methods) is answered only descriptively.

Table 15. Where each sub-question and each Phase 2 gap question is answered.

Question	Evidence	Answer	Where
SQ1: what fraction of "forgotten" models recover under each condition?	Verdict-flip rate among K-forgotten checkpoints	C1-low 0/13; C1-high 4/13 (30.8%); C2 1/13; C3 0/13; C4 1/17 (ROUGE-L)	Table 7; Figures 3–5
SQ2: how well does each pre-attack metric predict recovery?	Spearman ρ with G; AUROC	Memorization metrics: ρ 0.00 to 0.10 under C1-high. Token entropy −0.74, Truth Ratio −0.67. Checkpoint-level AUROC not computed (4 recovered). Item-level AUROC 0.355–0.426 for memorization metrics.	Table 9; Figures 6 and 8; Section 6
SQ3: do the answers change across methods and configurations?	Results by method; leave-one-method-out	Yes. RMU: 2/2 flips and 3/6 recovered. Leaving one method out moves the flip rate between 18.2% and 42.9%.	Tables 10–11
Gap question 1: do paraphrase and mixed prompts add failures beyond relearning and quantization?	Ablation A3	No. They uniquely catch no K-forgotten checkpoint by answer probability.	Table 13
Gap question 2: do metrics that predict relearning recovery also predict recovery under the other conditions?	Spearman ρ per condition	Not answerable positively: the memorization metrics did not predict relearning recovery, and the other conditions produced almost no recovery to predict.	Table 9; Section 5.4
Gap question 3: do the metric rankings of [3] hold across methods and configurations?	Flips per metric; comparison with [3]	Answered descriptively only: Extraction Strength, ranked highest in [3], marks the most checkpoints as forgotten and is the only metric whose verdicts paraphrasing overturns.	Tables 8 and 12

9.2 Strengths

- Every condition is applied to the same checkpoints and to a retain-only control, so recovery is separated from what the condition teaches by itself.
- Relearning and evaluation use disjoint authors, and ablation A5 shows that recovery on the held-out authors is as large as on the relearning authors.
- The checkpoint list, the split, the thresholds and the selection rule were fixed and committed before the runs, and every result reported in this paper is produced by a script from the saved result files.

9.3 Limitations

- One benchmark (TOFU forget10), one model size (1B) and 24 checkpoints. Checkpoints of the same method are not independent. Intervals are wide and describe this pool only.
- No checkpoint met the pre-set forgotten verdict, so flip rates refer to $K \leq 0.10$ without the utility requirement.
- Utility is a proxy measured on 200 retain questions and is not comparable with published TOFU Model Utility. Forget Quality was not computed.
- Relearning uses LoRA with one seed, so recovery may be lower than with full fine-tuning and its run-to-run variation is unknown.
- Quantization is simulated, its group size is our own choice, and it damages the 1B model heavily, which limits what C2 can show.
- C4 is measured with ROUGE-L only, and ROUGE-L separates the original and control models poorly under C2–C4.
- Of the 200 paraphrases, 20 were checked by hand; some of the others add a word such as "literary".
- Generated outputs were not saved, so they could not be read, grouped by question type or checked for refusals.

10. Conclusion

We evaluated 24 released unlearned checkpoints under five condition settings with a retain-only control. None achieved forgetting while keeping 90% of utility. Of the 13 that met the forgetting threshold alone, about one third recovered the knowledge after fine-tuning on 40 examples about other authors, and the scores usually used to declare forgetting did not indicate which ones would. A forgetting claim based on a single low score should therefore be accompanied by a relearning test at an adequate budget and by checks of utility and output entropy. Future work should repeat the study with several seeds, full fine-tuning, larger models and a second benchmark such as MUSE-News.

Reproducibility

The complete implementation is in a public repository: https://github.com/fizzahussain/LLM-Unlearning-Research. It contains the configuration file, the frozen checkpoint manifest, the author split, the 200 paraphrases with their prompt and review record, the mixed prompts, the run script (scripts/run_checkpoints.py), the 156 per-question result files of the main experiment, the 78 result files of ablation A5 and the run logs. Running scripts/analyze_results.py reproduces every table in this paper from the result files, and scripts/make_figures.py reproduces every figure. The commit history records that the manifest, split and thresholds were fixed before the runs.

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Reference numbers are those used in the Phase 1 and Phase 2 reports, so that citations stay consistent across the three phases; only the works cited in this paper are listed.

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Appendix A. Results for Each Checkpoint

Table A1. Residual knowledge K and utility before any condition, and control-adjusted recovery G under each condition, for the 24 checkpoints.

Checkpoint	K at C0	Utility / original	K-forgotten	G C1-low	G C1-high	G C2	G C3	G C4 (ROUGE-L)
GradDiff 1	−0.01	0.52	Yes	0.04	0.66	0.29	−0.05	−0.03
GradDiff 2	−0.16	0.46	Yes	−0.01	0.06	0.03	0.03	0.10
GradDiff 3	−0.16	0.46	Yes	−0.01	−0.08	0.03	0.03	0.06
GradDiff 4	−0.16	0.82	Yes	−0.01	−0.07	0.03	0.03	−0.01
GradDiff 5	−0.16	0.86	Yes	−0.01	−0.08	0.03	0.03	0.01
GradDiff 6	−0.16	0.81	Yes	−0.01	−0.08	0.03	0.03	−0.01
NPO 1	−0.08	0.14	Yes	0.09	0.31	0.13	0.02	−0.03
NPO 2	0.26	0.70	No	0.00	0.04	0.23	−0.03	−0.04
NPO 3	−0.14	0.75	Yes	0.01	0.15	0.03	0.02	−0.19
NPO 4	0.39	0.99	No	0.00	−0.01	−0.17	−0.10	−0.07
NPO 5	−0.11	0.73	Yes	−0.01	−0.04	0.03	0.02	−0.31
NPO 6	0.24	0.90	No	−0.01	−0.04	−0.06	−0.05	−0.14
SimNPO 1	0.94	1.03	No	−0.01	−0.06	−0.45	−0.35	−0.55
SimNPO 2	0.94	1.04	No	−0.01	−0.06	−0.45	−0.35	−0.50
SimNPO 3	0.38	1.04	No	0.05	0.21	−0.08	−0.11	−0.09
SimNPO 4	0.14	1.00	No	0.04	0.32	0.02	−0.04	−0.02
SimNPO 5	−0.09	0.80	Yes	−0.00	0.04	0.04	0.03	−0.07
SimNPO 6	−0.08	0.89	Yes	−0.00	0.05	0.04	0.03	−0.02
RMU 1	0.93	0.92	No	0.02	−0.01	−0.70	−0.36	−0.58
RMU 2	0.94	0.89	No	−0.00	−0.05	−0.56	−0.35	−0.55
RMU 3	−0.12	0.17	Yes	0.05	0.68	−0.01	0.01	−0.02
RMU 4	0.90	0.90	No	−0.00	−0.04	−0.53	−0.33	−0.47
RMU 5	0.24	0.63	No	0.15	0.51	−0.30	−0.13	−0.03
RMU 6	−0.16	0.04	Yes	0.00	0.56	0.03	0.03	−0.01

K and G use answer probability except in the last column.

Table A2. Settings of the 24 released checkpoints, as encoded in their OpenUnlearning names.

Checkpoint	Learning rate	Epochs	Other settings
GradDiff 1	1e-05	5	alpha 1
GradDiff 2	2e-05	5	alpha 1
GradDiff 3	3e-05	5	alpha 1
GradDiff 4	3e-05	10	alpha 10
GradDiff 5	4e-05	10	alpha 10
GradDiff 6	5e-05	10	alpha 10
NPO 1	1e-05	5	alpha 1, beta 0.05
NPO 2	1e-05	10	alpha 5, beta 0.1
NPO 3	2e-05	10	alpha 2, beta 0.05
NPO 4	2e-05	5	alpha 2, beta 0.5
NPO 5	5e-05	5	alpha 1, beta 0.1
NPO 6	5e-05	10	alpha 5, beta 0.5

Checkpoint	Learning rate	Epochs	Other settings
SimNPO 1	1e-05	5	alpha 1, beta 3.5, gamma 0.125, delta 0
SimNPO 2	1e-05	10	alpha 1, beta 4.5, gamma 0.125, delta 0
SimNPO 3	2e-05	10	alpha 1, beta 3.5, gamma 0.25, delta 0
SimNPO 4	2e-05	5	alpha 1, beta 4.5, gamma 0.125, delta 1
SimNPO 5	5e-05	5	alpha 1, beta 3.5, gamma 0.25, delta 1
SimNPO 6	5e-05	10	alpha 1, beta 4.5, gamma 0.25, delta 1
RMU 1	1e-05	5	layer 5, steering coefficient 1
RMU 2	1e-05	10	layer 10, steering coefficient 100
RMU 3	2e-05	10	layer 5, steering coefficient 10
RMU 4	2e-05	5	layer 15, steering coefficient 10
RMU 5	5e-05	5	layer 10, steering coefficient 1
RMU 6	5e-05	10	layer 15, steering coefficient 100

All checkpoints are open-unlearning/unlearn_tofu_Llama-3.2-1B-Instruct_forget10_<method>_<settings>; the full identifiers are in manifests/checkpoint_manifest.csv.

Appendix B. Coverage of the Phase 3 Requirements

Table B1. Where each required component is reported.

Required component	Where in this paper
1. Implementation, feedback and documented changes	Section 3; Tables 1–2
2. Dataset, preprocessing, feature engineering, splits and setup	Section 4; Tables 3–4; Table A2
3. Baseline evaluation and metrics	Section 5.1; Tables 5–6; Figure 1
4. Graphical results	Figures 1–9
5. Comparison with existing research and fairness limits	Section 6; Table 12
6. Ablation studies	Section 7; Tables 13–14; Figures 8–9
7. Analysis, discussion, strengths, limitations and whether the gap is addressed	Sections 8–10; Table 15